Introduction

Harrell's concordance index (C-index) generalises the ROC AUC to censored survival data. It is the probability that, for a randomly chosen pair of subjects, the one with the higher model-predicted risk actually fails first. The C-index is therefore the headline measure of *discrimination* for any survival model — Cox regression, parametric survival, machine-learning predictors — and the default reporting standard alongside the model's coefficients in clinical prediction-model papers.

Prerequisites

A working understanding of survival analysis, Cox regression, censoring, and the AUC for binary classifiers; familiarity with the distinction between calibration and discrimination.

Theory

Define a pair (i, j) as *comparable* if it can be determined which subject failed first — at least one observed event with the other subject either failing later or surviving past the first failure. The C-index is the proportion of comparable pairs for which the higher predicted risk corresponds to the earlier observed event:

$$C = \mathrm{P}(\hat{\eta}_i > \hat{\eta}_j \mid T_i < T_j, \delta_i = 1).$$

$C = 0.5$ corresponds to random ranking and $C = 1.0$ to perfect discrimination. Variants address specific limitations of Harrell's original index: Uno's C-statistic uses inverse-probability-of-censoring weights for unbiased estimation under heavy or non-uniform censoring; time-dependent AUCs (Heagerty, Uno) describe how discrimination changes across the follow-up period; the integrated Brier score is a complementary calibration-discrimination compound.

Assumptions

Risk scores come from a survival model whose linear predictor $\hat{\eta} = X\hat{\beta}$ is a meaningful ranking; the censoring mechanism is non-informative for Harrell's C and may be informative for Uno's C up to its weighting.

R Implementation

```r
library(survival)

data(lung)
fit <- coxph(Surv(time, status) ~ age + sex + ph.ecog, data = lung)
summary(fit)$concordance

# Time-dependent c-index via survcomp or timeROC
```

Output & Results

`summary(fit)$concordance` returns Harrell's C-index together with its standard error. For richer diagnostics, `survcomp::concordance.index()` adds bootstrap confidence intervals and `timeROC::timeROC()` returns time-dependent AUCs at user-specified landmarks.

Interpretation

A reporting sentence: "The Cox model achieved a Harrell's C-index of 0.63 (95 % CI 0.56 to 0.70) on the development data, indicating modest discrimination; on the held-out validation cohort the C-index dropped to 0.59, suggesting some optimism that the bootstrap procedure did not fully correct." Always pair the development-set C with an external or cross-validated estimate; in-sample concordance is optimistic.

Practical Tips

- $C > 0.7$ is conventionally considered acceptable for clinical use, $C > 0.8$ good; thresholds depend on the application and on what alternative classifiers achieve.
- Use Uno's C-statistic (`survAUC::UnoC()`) when censoring is heavy or non-uniform; Harrell's C can be biased upward in those settings.
- Report bootstrap confidence intervals via `survcomp::concordance.index()`; the analytical SE in `coxph` output is conservative.
- Time-dependent AUCs (`timeROC::timeROC()`) complement the overall C-index by showing when discrimination is highest and lowest across follow-up.
- For external validation, compute the C-index on an independent dataset; a development-set C without validation should never be the only discrimination measure reported.
- Pair discrimination (C-index) with calibration (calibration-in-the-large, calibration slope, Brier score); a model can rank well but predict poorly calibrated absolute probabilities.

R Packages Used

`survival` for `coxph()` with built-in concordance computation; `survcomp` for `concordance.index()` and bootstrap CIs; `timeROC` for time-dependent AUCs; `pec` for prediction-error curves and integrated Brier scores; `riskRegression` for unified discrimination and calibration metrics.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Survival primer: KM, log-rank, Cox PH
- Time-varying covariates and immortal-time bias
- Competing risks and multistate models
- Survival ML (random survival forest, DeepSurv)
- Time-dependent Brier, IPA, external validation